Theory case · Topic 2 · 2 points · 15 min

C18 — A launcher maker buys its engine supplier: vertical integration

Helios Launch S.A. builds medium rockets. Its engines come from a single independent supplier that has raised prices twice in two years and delayed deliveries, and the two firms argue about who must pay for each design change. Helios is negotiating to buy the supplier. A director objects: engine technology is changing fast, the supplier also sells engines to other launch companies, and Helios has no experience in managing a factory of that size. Another director proposes, instead, to also buy the satellite-integration company that prepares customers' payloads before launch.

Part (a) · 1.25 points

Identify the type of vertical integration in each proposal and the advantages Helios seeks with the purchase of the engine supplier

The deck names forward and backward vertical integration [Slide 27]; in the standard reading (extension), backward integration moves toward the inputs and forward integration toward the customer. Buying the engine supplier is therefore **backward vertical integration**, and buying the satellite-integration company would be **forward vertical integration**. Both change the **vertical scope** of the firm, replacing a market interface with an administrative one [Slide 19].

The advantages Helios seeks with the engine supplier [Slide 27]:

- **control of inputs**: no more unexpected price rises or delays;
- **reduction of transaction costs**: the disputes over design changes become internal decisions;
- **independence** from a single supplier and a **better competitive position**;
- **reduction of information asymmetries**: Helios will know the real costs and status of the engines.

KEY POINTS

- Buying the engine supplier: backward vertical integration; buying payload integration: forward vertical integration.
- Advantages: control of inputs, reduction of transaction costs (disputes over design changes), independence, better competitive position, reduction of information asymmetries.

Part (b) · 0.75 points

Use the disadvantages of vertical integration to evaluate the director's objection

The director's points match the **disadvantages** of vertical integration [Slide 27]. When engine technology changes fast, owning the factory means **decreased flexibility**: Helios cannot simply switch to a better engine on the market. If rival launch companies stop buying from a supplier owned by a competitor, the factory loses volume and the **advantages of specialization**. Running a large factory without experience brings **increased administrative costs** and **increased risk**. A balanced conclusion: integration is worth it if engines are critical and the supplier abuses its power, but Helios should first consider alternatives such as a long-term contract or a second supplier.

KEY POINTS

- Decreased flexibility: tied to its own engine technology when it changes fast.
- Loss of specialization advantages: the supplier's scale from selling to other launchers may be lost.
- Increased administrative costs and risk: managing a large new factory without experience.